

Jackfruit Phase-1

**Software
Specification**

Requirements

for

**Blood Bank Management
System**

Version 1.0 approved

Team No.: 9

Team Members:

Shreelakshmi G Bhat - SRN: PES2UG24CS478

Sangadala Yeasaswani - SRN: PES2UG24CS441

Shreyash M Yadwad- SRN: PES2UG24CS493

Shreenandan Jamadar - SRN: PES2UG24CS479

Table of Contents

Table of Contents.....	1
1.1 Purpose.....	2
1.2 Intended Audience and Reading Suggestions.....	3
1.3 Product Scope.....	3
1.4 References.....	3
2. Overall Description.....	3
2.1 Product Perspective.....	3
2.2 Product Functions.....	3
2.3 User Classes and Characteristics.....	4
Donor.....	4
Blood Bank Staff.....	4
Hospital / Patient (Recipient).....	4
Administrator.....	4
2.4 Operating Environment.....	4
2.5 Design and Implementation Constraints.....	5
2.6 Assumptions and Dependencies.....	5
3. External Interface Requirements.....	6
3.1 User Interfaces.....	6
3.2 Software Interfaces.....	6
3.3 Communications Interfaces.....	6
4. Analysis Models.....	6
5. System Features.....	8
5.1 User Management.....	8
5.1.1 Description and Priority.....	8
5.1.2 Stimulus/Response Sequences.....	8
5.1.3 Functional Requirements.....	8
5.2 Donor Management.....	8
5.2.1 Description and Priority.....	8
5.2.2 Stimulus/Response Sequences.....	9
5.2.3 Functional Requirements.....	9
5.3 Inventory Management.....	9
5.3.1 Description and Priority.....	9
5.3.2 Stimulus/Response Sequences.....	9
5.3.3 Functional Requirements.....	9
5.4 Blood Request & Availability.....	10
5.4.1 Description and Priority.....	10
5.4.2 Stimulus/Response Sequences.....	10
5.4.3 Functional Requirements.....	10
5.5 Donation Camp / Event Management.....	11
5.5.1 Description and Priority.....	11
5.5.2 Stimulus/Response Sequences.....	11

5.5.3 Functional Requirements.....	11
5.6 Notifications & Alerts.....	11
5.6.1 Description and Priority.....	11
5.6.2 Stimulus/Response Sequences.....	12
5.6.3 Functional Requirements.....	12
5.7 Reporting & Analytics.....	12
5.7.1 Description and Priority.....	12
5.7.2 Stimulus/Response Sequences.....	12
5.7.3 Functional Requirements.....	12
5.8 Search & Matching.....	12
5.8.1 Description and Priority.....	12
5.8.2 Stimulus/Response Sequences.....	12
5.8.3 Functional Requirements.....	13
5.9 Admin Functions.....	13
5.9.1 Description and Priority.....	13
5.9.2 Stimulus/Response Sequences.....	13
5.9.3 Functional Requirements.....	13
5.10 Certificate / Acknowledgment.....	13
5.10.1 Description and Priority.....	13
5.10.2 Stimulus/Response Sequences.....	14
5.10.3 Functional Requirements.....	14
6. Other Nonfunctional Requirements.....	15
6.1 Performance Requirements.....	15
6.2 Safety Requirements.....	15
6.3 Security Requirements.....	16
6.4 Software Quality Attributes.....	16
6.5 Business Rules.....	18
7. Other Requirements.....	18
Appendix A: Glossary.....	19
Appendix B: Field Layouts.....	19
Appendix C: Requirement Traceability Matrix.....	19

1. Introduction

1.1 Purpose

This document specifies the software requirements for Version 1.0 of the Blood Bank Management System (BBMS), a web-based application to manage blood donation and availability across blood banks. This SRS covers the complete system: donor and staff-facing functionality, blood inventory management, blood requests, donation camps, notifications, reporting, and administration.

1.2 Intended Audience and Reading Suggestions

This document is intended for the development team (Team 9), the course instructor/evaluator, and any future developers or testers who extend the system. Readers unfamiliar with the project should first read Section 2 (Overall Description) for context, then Section 5 (System Features) for detailed functional requirements, and Section 6 for non-functional, safety, and security requirements. Appendices provide supporting reference material (glossary, field layouts, and traceability).

1.3 Product Scope

The Blood Bank Management System digitizes and streamlines the process of managing blood donations, donor eligibility, blood inventory, and requests from hospitals or patients. Its purpose is to reduce blood wastage through expiry and stock tracking, speed up matching of compatible blood during emergencies, and give administrators full visibility and control across one or more blood bank branches. The product will be built using Python (Django) and/or JavaScript (Node.js).

1.4 References

- IEEE-style Software Requirements Specification template (“Jackfruit Phase-1”), as provided for this course.
- Course material: Software Engineering, UE23CS341A — Unit 1, Lecture 8: “Requirements Engineering: Functional, Non-Functional, Security Requirements.”
- Team 9 — “Updated Blood Bank Requirements (Final)” — finalized functional and non-functional requirements list used as the basis for this SRS.

2. Overall Description

2.1 Product Perspective

The BBMS is a new, self-contained web application; it is not a follow-on to an existing system. It may optionally integrate with external hospital or healthcare management systems via APIs (see NFR18) and with third-party SMS/email gateways for notifications (see NFR19). The system is composed of a backend (Django and/or Node.js), a relational database, and a responsive web frontend accessed by four classes of users.

2.2 Product Functions

At a high level, the system shall let users:

- Register and log in with role-based access (donor, staff, hospital, admin).
- Register as a donor, track eligibility, and view donation history.
- Manage blood inventory: add/update/remove units, track expiry, and view real-time stock.
- Search for and request blood, with compatibility checks and reservation.
- Schedule and participate in donation camps.
- Receive notifications and emergency broadcasts.
- View reports and dashboards on donations, stock, and requests.
- Search for suitable donors or nearby blood banks.
- Administer blood banks, staff, and user access, with full audit logging.
- Automatically receive a donation certificate/acknowledgment after donating.

2.3 User Classes and Characteristics

Donor

General public members who register, track eligibility, and donate. Low-to-moderate technical expertise expected; the interface must be simple and require minimal training (see NFR12).

Blood Bank Staff

Manage day-to-day inventory, donor records, camps, and process incoming requests. Moderate technical expertise; frequent daily use.

Hospital / Patient (Recipient)

Search for and request blood, including emergency requests. May use the system infrequently but during time-critical situations, so performance and clarity are important.

Administrator

Manage blood banks, staff accounts, access, and approvals; view audit logs and system-wide reports. Highest privilege level; expected to have the most familiarity with the system.

2.4 Operating Environment

- Web-based application accessible via major browsers: Chrome, Firefox, Edge, Safari.
- Backend: Python (Django) and/or JavaScript (Node.js); a relational database (e.g., PostgreSQL/MySQL).
- Deployable on supported cloud platforms (e.g., AWS, Heroku) or on-premise infrastructure, with optional Docker-based containerized deployment (see NFR23, NFR24).

- Client devices: desktop and mobile, given the responsive-design requirement (see NFR14).

2.5 Design and Implementation Constraints

- The technology stack is fixed to Python (Django) and/or JavaScript (Node.js), as mandated by the project brief.
- All access must go through role-based access control (RBAC) and HTTPS/token-based authentication (see NFR6, NFR7).
- The system must be developed within the academic semester timeline by a team of 4 members.
- Operational parameters (minimum donor eligibility interval, stock thresholds, user roles, notification channels, etc.) must be configurable by administrators rather than hard-coded.

2.6 Assumptions and Dependencies

- Users have access to a stable internet connection to use the web-based system.
- Blood bank staff will receive basic training on using the system before deployment.
- Third-party SMS/email gateway services are available and reliable for notifications.
- The initial deployment targets a single country/region's regulatory requirements.
- Integration with external hospital systems (NFR18) depends on those systems exposing compatible APIs; this is treated as optional/future scope.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide role-specific web dashboards (Donor, Staff, Hospital, Admin) with a consistent visual style, standard navigation, and standard error-message presentation. Common elements (login/logout, notifications, help) shall appear consistently across screens. The interface shall be responsive for both desktop and mobile use (see NFR14). Detailed UI mockups are to be maintained in a separate UI specification/wireframe document.

3.2 Software Interfaces

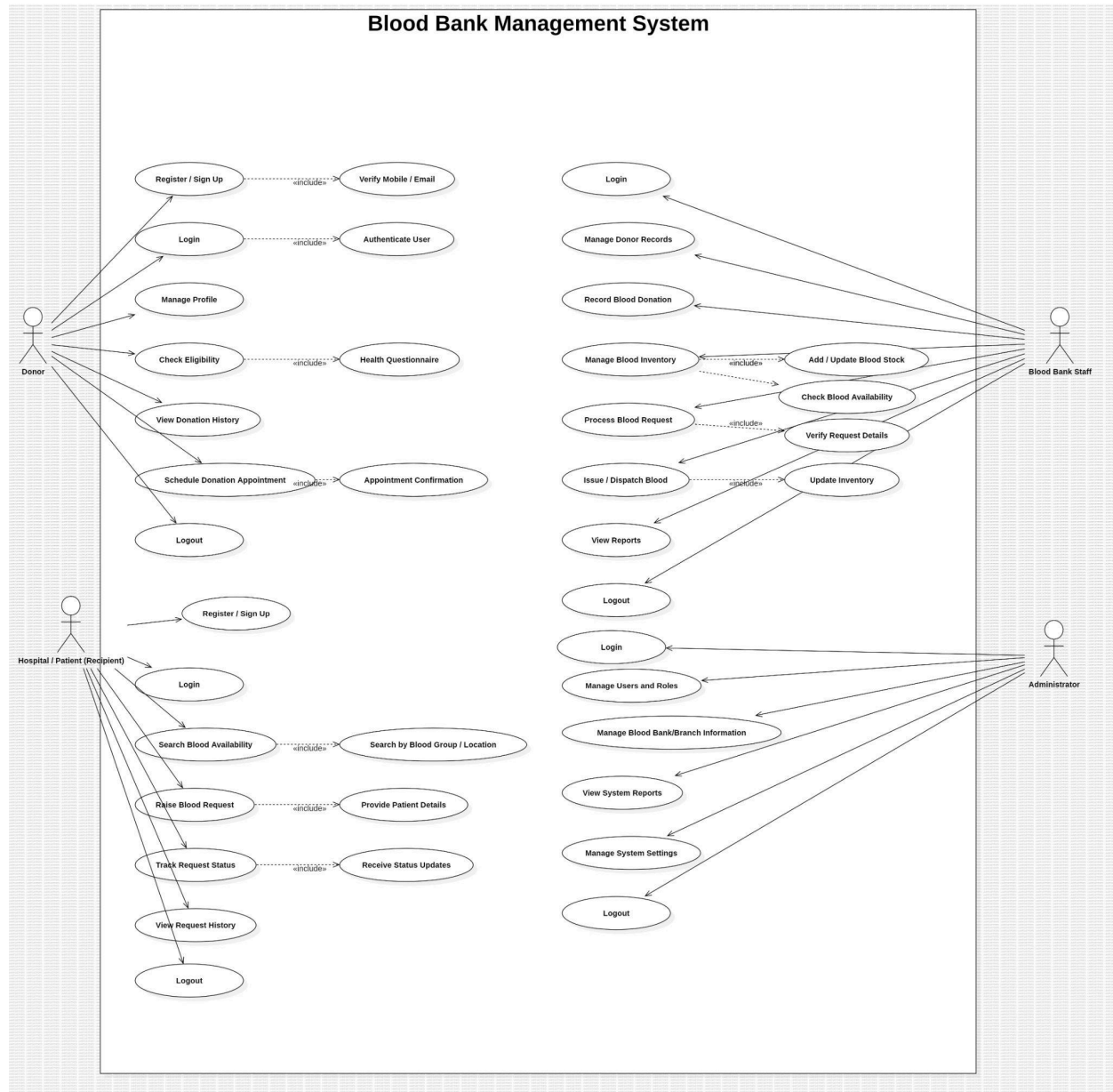
- Database: A relational database (e.g., PostgreSQL/MySQL) storing users, donors, inventory, requests, camps, and audit logs.
- Backend framework: Django (Python) and/or Node.js (JavaScript), exposing REST APIs consumed by the web frontend.
- SMS/Email gateway: Third-party service (e.g., Twilio, SendGrid, or equivalent) for notifications and emergency broadcasts (FR20, FR21, NFR19).
- Optional external hospital/healthcare management systems, integrated via REST APIs (NFR18).

3.3 Communications Interfaces

All client–server communication shall use HTTPS. The system shall expose REST APIs using JSON as the data-interchange format. Token/session-based authentication (e.g., JWT) shall be used to secure API communications (see NFR7). Notification delivery shall use standard email (SMTP/API-based) and SMS gateway protocols.

4. Analysis Models:

Use case diagram:



The use case diagram illustrates the interactions between four primary actors — Donor, Hospital/Patient (Recipient), Blood Bank Staff, and Administrator — and the major functions provided by the Blood Bank Management System. Donors primarily interact with registration, login, profile management, eligibility checking, donation history, and appointment scheduling, while hospitals and patients interact with blood availability search, request raising, request tracking, and request history. Blood Bank Staff manage donor records, record donations, maintain blood inventory, process incoming requests, and dispatch blood, while Administrators oversee user and role management, branch information, system-wide reports, and system settings.

Several core actions are supported by underlying verification and confirmation steps, modeled as «include» relationships, since these steps are mandatory whenever the parent action occurs. Registration always includes mobile/email verification, and login always includes user authentication, ensuring identity is confirmed before access is granted. Eligibility checking includes a health questionnaire, and appointment

scheduling includes appointment confirmation, supporting safe and reliable donation planning. On the recipient side, searching for blood availability includes filtering by blood group and location, while raising a request includes capturing patient details, and tracking a request includes receiving status updates — keeping recipients informed throughout the process. On the staff side, inventory management includes both stock updates and availability checks, request processing includes verifying request details, and dispatching blood includes updating inventory, ensuring that stock levels stay accurate and consistent with every transaction. Together, these included steps act as built-in safeguards and data-integrity checks that run automatically underneath the system's primary donor, recipient, and staff workflows.

5. System Features

Functional requirements are organized below by system feature (major service area), matching the finalized requirements list for this project. Each feature lists its description/priority, a representative stimulus/response sequence, and its itemized functional requirements (FR IDs).

5.1 User Management

5.1.1 Description and Priority

Registration, login, profile management, and account security for all user roles (Donor, Blood Bank Staff, Hospital Staff, Administrator). Overall feature priority: High.

5.1.2 Stimulus/Response Sequences

A user navigates to the login/registration screen, enters credentials or registration details, and the system authenticates the user or creates a new account, then redirects to a role-specific dashboard.

5.1.3 Functional Requirements

ID	Priority	Requirement Description
FR1	High	The system shall allow donors, blood bank staff, hospital staff, and administrators to register and log in using role-based access control.
FR2	High	The system shall allow users to manage profiles, including personal details, blood group, medical information, and contact information.
FR3	High	The system shall provide password reset and OTP-based verification for account security.

5.2 Donor Management

5.2.1 Description and Priority

Donor registration, eligibility tracking, and donation history management. Overall feature priority: High.

5.2.2 Stimulus/Response Sequences

A donor logs in and views or updates their health and donation details; the system validates their eligibility and displays the next eligible donation date.

5.2.3 Functional Requirements

ID	Priority	Requirement Description
FR4	High	The system shall allow donors to register and record blood group, age, weight, and health eligibility information.
FR5	High	The system shall maintain a complete donation history for each donor.
FR6	High	The system shall validate donor eligibility based on the configured minimum interval between donations and other defined eligibility criteria.
FR7	Medium	The system shall calculate and display the next eligible donation date and provide reminders when appropriate.

5.3 Inventory Management

5.3.1 Description and Priority

Blood-unit record management, expiry tracking, stock visibility, traceability, and disposal. Overall feature priority: High.

5.3.2 Stimulus/Response Sequences

Authorized staff add, update, or remove a blood-unit record; the system validates and stores it with a unique ID, expiry date, and location, and the inventory dashboard reflects the updated stock in real time.

5.3.3 Functional Requirements

ID	Priority	Requirement Description
FR8	High	The system shall allow authorized staff to add, update, and remove blood-unit records using blood group, quantity, collection date, and blood bank location.

ID	Priority	Requirement Description
FR9	High	The system shall track expiry dates and automatically flag expired blood units so they cannot be issued.
FR10	High	The system shall provide a real-time view of blood-stock levels by blood group and blood bank location.
FR11	High	The system shall generate low-stock alerts for configured blood groups and notify authorized administrators.
FR30	High	The system shall maintain a unique identification number for each blood unit and allow authorized staff to trace its collection, storage, allocation, and disposal history.
FR33	Medium	The system shall allow authorized staff to record disposal of expired, damaged, or unusable blood units and the reason for disposal.
FR34	Medium	The system shall allow authorized staff to filter and sort inventory by blood group, expiry date, collection date, and blood bank location.

5.4 Blood Request & Availability

5.4.1 Description and Priority

Searching for blood, submitting and processing requests, compatibility verification, and reservation. Overall feature priority: High.

5.4.2 Stimulus/Response Sequences

A hospital or patient searches for blood by group and location and submits a request; staff review and approve, reject, or fulfill the request, and the system reserves and tracks its status until fulfilled or cancelled.

5.4.3 Functional Requirements

ID	Priority	Requirement Description
FR12	High	The system shall allow authorized hospitals and patients to search for available blood by blood group and location.
FR13	High	The system shall allow users to submit blood requests and specify whether each request is emergency or scheduled.
FR14	High	The system shall allow blood bank staff to approve, reject, or fulfill blood requests.

ID	Priority	Requirement Description
FR15	Medium	The system shall search other participating blood banks when the required blood is unavailable at the selected/local blood bank.
FR16	High	The system shall verify blood-group compatibility before a blood request is fulfilled.
FR31	High	The system shall temporarily reserve compatible blood units for an approved blood request until the request is fulfilled or cancelled.
FR32	High	The system shall allow authorized hospitals and patients to track request status, including pending, approved, rejected, fulfilled, or cancelled.
FR35	High	The system shall validate blood requests by checking required information, blood availability, compatibility, and requested quantity before approval.

5.5 Donation Camp / Event Management

5.5.1 Description and Priority

Scheduling, publishing, and running blood donation camps and drives. Overall feature priority: Medium.

5.5.2 Stimulus/Response Sequences

Staff create and publish a donation camp; donors register for it; after the camp, staff record the collected units and the system automatically updates inventory.

5.5.3 Functional Requirements

ID	Priority	Requirement Description
FR17	Medium	The system shall allow authorized staff to schedule and publish blood donation camps and drives.
FR18	Medium	The system shall allow eligible donors to register for participation in donation camps.
FR19	High	The system shall update blood inventory with successfully collected units after a donation camp.

5.6 Notifications & Alerts

5.6.1 Description and Priority

Automated SMS/email notifications and emergency broadcasts. Overall feature priority: Medium.

5.6.2 Stimulus/Response Sequences

A triggering event occurs (eligibility, low stock, emergency, or request-status change); the system composes and sends an SMS/email or broadcast message to the relevant users.

5.6.3 Functional Requirements

ID	Priority	Requirement Description
FR20	Medium	The system shall send SMS/email notifications for donation eligibility, camp announcements, urgent requirements, and request-status updates.
FR21	Medium	The system shall support emergency broadcast messages to eligible nearby donors.

5.7 Reporting & Analytics

5.7.1 Description and Priority

Reports and dashboards on donations, stock, requests, and wastage. Overall feature priority: Medium.

5.7.2 Stimulus/Response Sequences

An admin selects a report type or date range; the system aggregates the relevant data and displays a report or dashboard with statistics and charts.

5.7.3 Functional Requirements

ID	Priority	Requirement Description
FR22	Medium	The system shall generate reports on donations collected, blood-group-wise stock, fulfilled/pending requests, and wastage from expired units.
FR23	Medium	The system shall provide an administrative dashboard with relevant statistics and charts.

5.8 Search & Matching

5.8.1 Description and Priority

Finding suitable donors or nearby blood banks/hospitals for a required blood type. Overall feature priority: High.

5.8.2 Stimulus/Response Sequences

A user enters blood-group and location criteria; the system matches these against donor or inventory records and returns filtered results.

5.8.3 Functional Requirements

ID	Priority	Requirement Description
FR24	High	The system shall allow authorized users to search for suitable donors by blood group and location for emergency requests.
FR25	Medium	The system shall allow users to search for nearby blood banks or hospitals that may hold the required compatible blood type.

5.9 Admin Functions

5.9.1 Description and Priority

Administration of blood banks, hospitals, staff, access, registrations, and audit logs. Overall feature priority: High.

5.9.2 Stimulus/Response Sequences

An admin logs into the admin console, manages branches, staff, or access, or reviews pending registrations; the system applies the changes and records the action in the audit log.

5.9.3 Functional Requirements

ID	Priority	Requirement Description
FR26	High	The system shall allow administrators to manage blood banks, hospitals, staff accounts, and user access.
FR27	High	The system shall allow administrators to approve and verify donor and hospital registrations.
FR28	High	The system shall maintain audit logs of important system transactions.

5.10 Certificate / Acknowledgment

5.10.1 Description and Priority

Automatic generation of donation certificates or acknowledgments. Overall feature priority: Low.

5.10.2 Stimulus/Response Sequences

A donation is marked complete; the system automatically generates a certificate/acknowledgment, which the donor is notified of and can download.

5.10.3 Functional Requirements

ID	Priority	Requirement Description
FR29	Low	The system shall automatically generate a donation certificate or acknowledgment for a donor after a successful donation.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

The following requirements ensure the system remains responsive under normal and peak (emergency/camp) load.

ID	Priority	Requirement Description
NFR1	High	Blood availability search and query results should normally be returned within 2–3 seconds.
NFR2	High	The system shall support concurrent access by multiple users during emergencies and donation camps while maintaining the defined performance requirements.

6.2 Safety Requirements

The following requirements address possible harm that could result from incorrect or unsafe use of the product — specifically, the risk of an incompatible or unsafe blood unit being issued. These are derived from, and cross-referenced to, related functional and non-functional requirements above.

ID	Priority	Requirement Description
SAFETY-1	High	The system shall block the fulfillment of any blood request unless blood-group compatibility has been verified (see FR16, FR35), to prevent transfusion of incompatible blood.
SAFETY-2	High	The system shall prevent expired blood units from being issued or allocated to any request (see FR9), to avoid harm from administering unsafe blood units.
SAFETY-3	High	The system shall prevent the same blood unit from being allocated to more than one fulfilled request (see NFR20), to avoid double-issuing a unit already used elsewhere.
SAFETY-4	Medium	The system shall provide recovery/failover mechanisms for critical modules such as inventory and blood requests (see NFR11), to avoid loss of access to life-critical information during emergencies.

6.3 Security Requirements

The following requirements address authentication, authorization, and protection of sensitive donor and medical data.

ID	Priority	Requirement Description
NFR5	High	Sensitive personal and medical information shall be protected using appropriate encryption and secure password storage.
NFR6	High	Role-based access control (RBAC) shall restrict features and data according to user roles.
NFR7	High	API endpoints shall use HTTPS and secure token/session-based authentication where applicable.
NFR8	High	The system shall comply with applicable data-protection and privacy regulations.

6.4 Software Quality Attributes

Additional quality characteristics important to users and developers, grouped by attribute.

Scalability

ID	Priority	Requirement Description
NFR3	Medium	The system should support expansion from a single blood bank to multiple branches or a larger network.
NFR4	Medium	The database should efficiently handle increasing donor, inventory, request, and transaction records while maintaining acceptable response time.

Reliability & Availability

ID	Priority	Requirement Description
NFR9	High	The system should target 99.5% or higher availability, particularly for critical blood-request functions.
NFR10	High	The system should perform automated backups of donor, inventory, request, and transaction data.
NFR11	Medium	The system should provide recovery/failover mechanisms for critical modules such as inventory and blood requests.

Usability

ID	Priority	Requirement Description
NFR12	High	The system shall provide a simple and intuitive interface that allows non-technical donor users to complete common tasks with minimal training.
NFR13	Medium	The system should support multilingual content where required to reach a wider donor base.
NFR14	High	The web interface should be responsive and usable on desktop and mobile devices.

Maintainability

ID	Priority	Requirement Description
NFR15	Medium	The system should use a modular architecture with separate modules for users, donors, inventory, requests, notifications, and reporting.
NFR16	Medium	APIs and major modules shall be documented to support maintenance and future development.
NFR17	Medium	The codebase shall use version control, with automated build/test/deployment practices where applicable.

Interoperability

ID	Priority	Requirement Description
NFR18	Low	The system should provide APIs that can integrate with compatible external hospital or healthcare management systems.
NFR19	Medium	The system should support integration with SMS/email gateways for notifications.

Portability

ID	Priority	Requirement Description
NFR23	Low	The system should be deployable on supported cloud platforms or on-premise infrastructure.

ID	Priority	Requirement Description
NFR24	Low	The application should support containerized deployment using Docker for consistent deployment across environments.

6.5 Business Rules

Operating principles governing how the system behaves, including domain requirements arising from the blood-banking domain itself.

ID	Priority	Requirement Description
NFR20	High	The system shall prevent the same blood unit from being allocated to more than one fulfilled request.
NFR21	High	Inventory updates shall remain transactionally consistent when multiple requests are processed concurrently.
NFR22	High	The system shall maintain logs of donations, requests, inventory changes, approvals, and other critical actions with timestamps and responsible users.
BR-1	High	Blood shall only be allocated to a request when the donor/unit blood group is compatible with the recipient's blood group, per standard transfusion compatibility rules.
BR-2	High	A donor is only eligible to donate again after the configured minimum inter-donation interval has elapsed.
BR-3	Medium	The minimum donor eligibility interval, stock thresholds, user roles, notification channels, and other operational rules shall be configurable by authorized administrators rather than hard-coded.

7. Other Requirements

- **Legal/Regulatory:** The system shall comply with applicable healthcare data-protection and privacy regulations for donor and patient medical data (see NFR8).
- **Internationalization:** The system should support multilingual content to reach a wider donor base (see NFR13).
- **Reuse:** Where practical, the team will reuse established libraries/services for authentication (OTP/email/SMS gateways) rather than building these from scratch, to focus development effort on core blood-bank functionality.
- **Deployment:** The system should support containerized deployment (Docker) for consistent environments across development, testing, and (if applicable) production (see NFR24).

Appendix A: Glossary

Term	Definition
BBMS	Blood Bank Management System — the product specified in this document.
FR	Functional Requirement — a statement of a service the system must provide.
NFR	Non-Functional Requirement — a constraint or quality attribute the system must satisfy.
RBAC	Role-Based Access Control — restricting system features/data based on a user's assigned role.
OTP	One-Time Password — a temporary code used to verify a user's identity.
Blood Unit	A single collected and stored quantity of blood, uniquely identified and tracked through its lifecycle.
Eligibility Interval	The configured minimum time period a donor must wait between successive donations.
Compatibility	The rule set determining which donor blood groups may be safely given to which recipient blood groups.
SRN	Student Registration Number, used to identify each team member.

Appendix B: Field Layouts

Appendix C: Requirement Traceability Matrix